Bowen Xie

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RESEARCH INSTEERESTS

Applied Probability, Queueing Theory, Stochastic Control, Operations Research, Mathematical Finance, Reinforcement Learning, Numerical Analysis

EDUCATION

Doctor of Phiolosophy **Iowa State University**, Department of Mathematics Aug 2017 -Aug 2022
- Applied Mathematics, GPA: 3.92/4.00
- Advisor: Dr. Ananda Weerasinghe

Master of Science **Iowa State University**, Department of Mathematics Aug 2017 -May 2019
- Applied Mathematics, GPA: 3.90/4.00
- Advisor: Dr. Ananda Weerasinghe

Bachelor of Science **Liaoning Technical University**, Department of Mathematics Sep 2013 -July 2017
- Mathematics & Applied Mathematics, GPA: 3.51/4.00 (Math GPA: 3.77/4.00)

EXPERIENCE

The University of Akron, Assistant Professor — Akron, OH
Department of Mathematics, College of Engineering & Polymer Science — Fall 2024-present

Washington University in St. Louis, Postdoctoral Lecturer — St. Louis, MO
Department of Statistics and Data Science — Fall 2022-Spring 2024

Iowa State University, Graduate Teaching/Research Assistant — Ames, IA
Department of Mathematics — Fall 2017-Summer 2022

MENTORING AND STUDENTS

Niraj Jaishwal — Akron, OH
- Undergraduate student from Computer Science — Fall 2025-present

Samuel Adjei Baiden — Akron, OH
- Master student from Applied Mathematics — Fall 2024-Spring 2026

Benno Lutz — Akron, OH
- Undergraduate student honors project — Fall 2025-Spring 2026

PUBLICATIONS AND PREPRINTS

Publications

[15] Y. Bai, Y. Gao, B. Xie, and S. Luo, “The symmetry method for fractional airy equation,” *Mathematical Methods in the Applied Sciences*, vol. 49, no. 14, pp. 16 072–16 087, 2026.

[14] B. Xie and Y. Gao, “Numerical analysis for fractional riccati differential equations based on finite difference method,” *Journal of Nonlinear Modeling and Analysis https://journal. global-sci. org/jnma*, vol. 7, no. 1, pp. 189–208, 2025.

[13] B. Xie, “Multi-component matching queues in heavy traffic,” *Queueing Systems*, pp. 1–47, 2024, (Part of a Special Issue: Product Forms, Stochastic Matching, and Redundancy).

[12] B. Xie and Y. Gao, “Diffusion approximation of a special bandwidth sharing model via infinitesimal generators with a lifting-projection method,” *Operations Research Letters*, vol. 52, p. 107 063, 2024.

[11] B. Xie and Y. Gao, "On the long-run average cost minimization problem of the stochastic production-inventory model," *Journal of Industrial and Management Optimization*, vol. 20(5), pp. 1823–1844, 2024.

[10] B. Xie and Y. Gao, "Population growth model with fractional operator and its numerical analysis," *Journal of Mathematical Study*, 2024, (to be appeared).

[9] Y. Gao and B. Xie, "Numerical analysis for fractional bratu type equation with explicit and implicit methods," *Mathematical Methods in the Applied Sciences*, vol. 46, no. 17, pp. 18 447–18 457, 2023, (corresponding author).

Preprints and Working Papers

[8] S. Adjei Baiden, B. Xie, and Y. Gao, *Optimal dynamic pricing under double-ended reflected stochastic market potential*, (under review at IISE Transactions).

[7] B. Xie, J. Chávez-Casillas, and J. E. Figueroa-López, *Market making in limit order books*, (Working paper).

[6] S. Ray, R. Wu, and B. Xie, *Regulated multi-component matching queues with abandonment: Scaling limits and large deviations*, (near completion), 2026.

[5] B. Xie and Y. Gao, *Joint pricing and innovation control in regulated recycling-rate diffusion*, (submitted to IISE Transactions), 2026.

[4] B. Xie, *An infinitesimal generator approach on weak convergence of regulated multi-class matching systems*, arXiv preprint arXiv:2507.09789 (submitted to Journal of Applied Probability, R&R), 2025.

[3] B. Xie and H. Yin, *Admission control for a single server waiting time process in heavy traffic*, arXiv preprint arXiv:2212.05674 (to be submitted to Queueing Systems), 2025.

Theses

[2] B. Xie, "Topics of queueing theory in heavy traffic (ph. d thesis)," `https://dr.lib.iastate.edu/items/be350d6c-e38a-4793-8e40-33445c2cdb5f/full`, Ph.D. dissertation, 2022.

[1] B. Xie, "Long-run average cost minimization of a stochastic processing system (master creative component)," `https://dr.lib.iastate.edu/entities/publication/3f5cb727-8a92-46c2-8343-9252e1e7c78c`, M.S. thesis, 2019.

SELECTED RESEARCH PROJECTS

Dynamics of limit order books in high-frequency trading [7] — Fall 2023-present

Postdoc Mentor: Dr. José E. Figueroa-López, Washington University in St. Louis
Joint work with Dr. Jonathan Chávez Website, University of Rhode Island

- We aim to construct a matching system framework for LOBs and implement algorithms to address the optimal placement problem faced by market makers.
- Study the performance of reinforcement learning (RL) in solving market-making problems with inventory risk.
- Attempt to train market-making agent using RL with a real-world stock market constructed from historical Limit Order Book (LOB) data.
- Attempt to conduct Neural Networks to approximate state-value and state-action value functions and compare their performance with tile coding as well as the theoretical solution to the optimal placement problem.

Joint stochastic control on product and process innovations [5] — Spring 2025-present

- Production and process innovation are key drivers of quality improvement and cost reduction. In the circular economy, dynamic demand and resource recycling are crucial for operational efficiency.
- We explore stochastic control strategies within the propagation of chaos framework proposed in [5].
- The joint control of the product price and innovations under constraints such as recycling rates and demand propagation will be addressed.
- Our approach integrates rigorous stochastic analysis with the Actor-Critic

algorithm to develop adaptive and tractable solutions for complex systems.

- Applications in Monopoly's product and process innovations [8].

Scaling-limits on regulated matching systems [6] — Fall 2024-present

Joint work with Dr. Souvik Ray and Dr. Ruoyu Wu, Iowa State University

- Construct a diffusion approximation for a large-scale regulated matching system, characterized by a coupled regulated stochastic integral equation
- Provided a non-trivial approach to show the existence and uniqueness of the weak solution to such coupled regulated stochastic integral equations and simulated the matching behavior.
- Developing control strategies for heterogeneous matching systems, especially under resource constraints requiring admission or buffer policies.

Multi-component Matching Queue Systems in Heavy Traffic [13] — Summer 2019-2022

Advisor: Dr. Ananda Weerasinghe, Iowa State University

- Constructed many-component matching queue models and studied the diffusion-scaled heavy traffic limits, which are characterized by a coupled non-linear stochastic integral equation.
- Established the convergence of the cost functional under cost structures.
- Formulated an asymptotic relationship between the queue length processes and the virtual waiting time processes (the asymptotic Little's law) and further refined the result to L^p sense on a special probability space.
- Simulated the matching queue systems and the corresponding stochastic integral equations numerically in Python.
- Observe that a matching queue model without abandonment could possibly be interpreted as a basket option pricing problem.

INVITED TALKS

INFORMS Annual Meeting 2026 — San Francisco, CA
Scaling limits for regulated multi-component matching queues with abandonment — November 1-4, 2026

Stochastic Network Conference 2026, Poster presentation — Chicago Booth, IL
Scaling limits of regulated multi-component matching systems — June 14-19, 2026

SIAM Math Club Colloquium, at The University of Akron — Akron, OH
Inside High-Frequency Trading: The Mathematics of the Limit Order Book — Mar 13, 2026

INFORMS Annual Meeting 2025 — Atlanta, GA
From a single-server waiting time control to reinforcement learning motivation for matching systems — Oct 26-29, 2025

INFORMS Applied Probability Society Conference 2025 — Atlanta, GA
Admission control for a single server waiting time process in heavy traffic,
Intro to regulated matching systems — June 30-July 3, 2025

SIAM Math Club Colloquium, at The University of Akron — Akron, OH
Stochastic Control in Action: Theory, Applications, and Reinforcement Learning — Mar 21, 2025

Colloquium for Pure and Applied Math, at The University of Akron — Akron, OH
Queueing Theory and Stochastic Controls:
Applications in Industry, Economics, and Financial Math — Oct 11, 2024

2024 Joint Mathematics Meetings, AMS Special Session — San Francisco, CA
Dynamics of Matching Queues in Heavy Traffic and
Some New Findings of Regulated Matching Systems — Jan 3-6, 2024

2023 INFORMS Annual Meeting, Applied Probability Society Cluster — Phoenix, AZ
Multi-component matching and its dynamics and some new findings — Oct 15-18, 2023

Statistics and Data Science Seminar, at Washington University in St. Louis — St. Louis, MO
Many-component Matching Queue Systems in Heavy Traffic — Sep 21, 2022

SIAM Graduate Student Seminar, at Iowa State University — Ames, IA
Multi-component Matching Queues in Heavy Traffic — Apr 08, 2022

Probability and Statistics Seminar, at University of Wisconsin-Milwaukee — Virtual
Many-component Matching Queue System — Dec 1, 2021

20th Northeast Probability Seminar — Virtual
Many-component Matching Queue System — Nov 18, 2021

SIAM Graduate Student Seminar, at Iowa State University — Virtual
Optimal Control Problem for a Diffusion Process — Nov 06, 2020

COMMITTEES AND PROFESSIONAL SERVICE

INFORMS Education Outreach Graduate Committee — Akron, OH
Organize webinars and in-person panel discussions at the INFORMS Annual Meeting to promote graduate education. — Fall 2025-present

Reviewer, AMS Mathematical Reviews/MathSciNet Reviewer — Akron, OH
Reviewed scholarly papers for inclusion in the AMS Mathematical Reviews database and Mathematics Subject Classification. — Fall 2024-present

Reviewer, Methodology and Computing in Applied Probability — Akron, OH
Provide peer reviews for submitted manuscripts. — Fall 2024-present

Reviewer, SIAM Journal on Financial Mathematics — Akron, OH
Provide peer reviews for submitted manuscripts. — Fall 2024-present

Graduate Admission Committee, at The University of Akron — Akron, OH
Review graduate student applications and assistantship applications in Math Department. — Fall 2024-present

Administrative Assistant — Ames, IA
Assisted the Math Department with administrative tasks, including library organization, website updates, and the Math Genealogy Project. — Summer 2021

SIAM Student Chapter — Ames, IA
Served as treasurer and participated in the planning committee for the SIAM Graduate Student Seminar. — Fall 2020-Spring 2021

CONFERENCE ATTENDANCE

INFORMS Annual Meeting 2026, *San Francisco, CA* — November 1-4, 2026

Stochastic Network Conference 2026, *Chicago Booth, IL* — June 15-19, 2026

INFORMS Annual Meeting 2025, *Atlanta, GA* — Oct 26-29, 2025

SNAPP Summer School on Stein's Method for Queueing Approximations, *Virtual* — July 29-Aug 28, 2025

INFORMS APS Summer School, *Atlanta, GA* — June 25-29, 2025

INFORMS Applied Probability Society Conference 2025, *Atlanta, GA* — June 30-July 3, 2025

Joint Mathematics Meetings 2024, *San Francisco, CA* — Jan 2024

INFORMS Annual Meeting 2023, *Phoenix, AZ* — Oct 15-18, 2023

Seminar on Stochastic Processes, *Tucson, AZ* — Mar 8-12, 2023

First-Passage Percolation and Related Models Program, *ICTS, India (Hybrid)* — July 11-29, 2022

6th Annual Meeting of SIAM Central States Section, *Virtual* — Oct 2021

5th Annual Meeting of SIAM Central States Section, *Ames, IA* Oct 2019

41st Midwest Probability Colloquium, *Evanston, IL* Oct 2019

TEACHING EXPERIENCE

Teaching Experience at The University of Akron

Asst Professor, MATH 149 - Precalculus, *11 students*	Fall 2026
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, *44 students*	Fall 2026
Asst Professor, MATH 300 - Tools for Data Science, *16 students*	Spring 2026
Asst Professor, MATH 450/550 - Optimization, *8 undergrads, 5 grads*	Spring 2026
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, *42 students*	Fall 2025
Asst Professor, MATH 221 - Analytic Geometry-Calculus I, *34 students*	Fall 2025
Asst Professor, MATH 450/550 - Optimization, *9 students*	Spring 2025
Asst Professor, MATH 221 - Analytic Geometry-Calculus I, *40 students*	Spring 2025
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, *40 students*	Fall 2024
Asst Professor, MATH 221(Honors) - Analytic Geometry-Calculus I, *26 students*	Fall 2024

Teaching Experience at Washington University in St. Louis

Postdoctoral Lecturer, MATH 495/5070 - Stochastic Processes, *27 students*	Spring 2024
Postdoctoral Lecturer, MATH 493/5010 - Probability, *40 students*	Spring 2024
Postdoctoral Lecturer, MATH 5010 - Probability, *50 students*	Fall 2023
Postdoctoral Lecturer, MATH 493 - Probability, *8 students*	Summer 2023
Postdoctoral Lecturer, MATH 493 - Probability, *73 students*	Spring 2023
Postdoctoral Lecturer, MATH 3200 - Statistics and Data Analysis, *70 students*	Fall 2022
Postdoctoral Lecturer, MATH 493 - Probability, *48 students*	Fall 2022

Teaching Experience at Iowa State University

Lecturer, MATH 150 - Discrete Mathematics for Business and Social Sciences	Summer 2019
Recitation Instructor, MATH 143 - Preparation for Calculus	Spring 2021, Fall 2020, Fall 2019
Recitation Instructor, MATH 150 - Discrete Mathematics	Spring 2020
Recitation Instructor, MATH 151 - Calculus for Business and Social Sciences	Spring 2019
Graduate Teaching Assistant, MATH 645 - Advanced Stochastic Processes	Spring 2022
Graduate Teaching Assistant, MATH 481/581 - Numerical Methods for Diff Equ	Spring 2022
Graduate Teaching Assistant, MATH 517 - Finite Difference Methods	Spring 2022
Graduate Teaching Assistant, MATH 554 - Intro to Stochastic Processes	Fall 2021
Graduate Teaching Assistant, MATH 561 - Numerical Analysis	Fall 2021
Graduate Teaching Assistant, MATH 554 - Introduction to Stochastic Processes	Fall 2018
Graduate Teaching Assistant, MATH 414 - Analysis I	Spring 2022
Graduate Teaching Assistant, MATH 435 - Geometry I	Fall 2018
Graduate Teaching Assistant, MATH 373 - Introduction to Scientific Computing	Fall 2018
Graduate Teaching Assistant, MATH 104 - Introduction to Probability	Spring 2018
Graduate Teaching Assistant, MATH 304 - Combinatorics	Fall 2017
Graduate Teaching Assistant, MATH 201 - Introduction to Proofs	Fall 2017

GRADUATE COURSES

- Methods of Applied Math I, II
- Ordinary Differential Equation and Dynamic Systems
- Introduction to Stochastic Processes
- Advanced Stochastic Processes
- Machine Learning
- Numerical Analysis
- Numerical Linear Algebra
- Finite Difference Methods
- Advanced Special Topics: Continuous Time Asset Pricing Models
- Optimal Transport and Deep Learning
- Theory and Methods for Continuous Optimization (Audit)

OTHER PROJECTS

These projects are done for the laboratory parts of the Machine Learning course. Spring 2020

- **Naive Bayes Classifier for Text Classification.** Implement the Naive Bayes algorithm to tackle the "20 Newsgroups" classification problem.
- **Decision Tree Classifier.** Experiment with the Decision Tree classifier on the congressional voting records data set using Weka.
- **Neural Network Classifier.** Using the TensorFlow package to perform an experiment on the optical recognition of the handwritten digits data set.
- **Ensemble Learning.** Train ensemble models, random forest, and AdaBoost.M1, and construct an ensemble classifier by training 4 individual models on the blood transfusion service center data set.